Reproducing the Silver 2025 ML-Driven Strong-Lens Classifier: the Model 1a (HST-long) ResNet

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Internal Reproduction Report

Abstract

We reproduce the conventional-lens classification core of Silver et al. (2025): *Model 1a (HST-long)*, the simulated-HST ResNet trained on Einstein radii $0.5'' < \theta_{\rm E} < 1.5''$, whose published peak validation AUC is 0.9978 (their §3.2.1.1, reached by ∼epoch 150). We stand up the full `lenstronomy` simulation pipeline (SIE lens, Sérsic source, lens light + environmental galaxies, $0.031''$ pixels, Gaussian PSF), the paper's in-training noise augmentation layer ($\sigma_{\rm BKG} \sim \mathcal{U}(0, 0.2)$, $t_{\rm exp} \sim 10^{\mathcal{U}(2,6)}$ s, varied every iteration), and the Huang+2021 "shielded" ResNet (32-filter shields, 56,769 parameters) with the paper's exact optimizer schedule ($\mathrm{lr}_0 = 1.25 \times 10^{-3}$ decaying 5× every 80 epochs, 360 epochs, batch 64, Adam, BCE loss, 80/20 split). On a single NVIDIA L4 the 360-epoch run completes in ∼15 min and reaches a best validation $\mathbf{AUC = 0.9937}$ — within $\mathbf{0.4}$ % of the published 0.9978 — confirming the headline "near-100 % completeness and purity for conventional lenses" claim with an order-of-magnitude smaller (4,000-image) starter set and analytic-Sérsic sources standing in for the paper's VELA hydrodynamical source stamps. The JWST regimes (Models 2/3), HST-short (Model 1b), and the small-$\theta_{\rm E}$ U-Net pixel localizer are explicitly out of scope.

Contents

1 Introduction

Silver et al. (2025) extend the strong-lens discovery space down to very small Einstein radii ($\theta_{\mathrm{E}} \lesssim 0.03''$) and very low halo mass ($M_{\mathrm{halo}} < 10^{11}\,M_{\odot}$) by combining JWST-resolution simulations with machine learning. Their pipeline forecasts detectable lenses from the CosmoDC2 (Korytov et al., 2019) lens catalog plus a source catalog to 29th magnitude, simulates lensed/unlensed images with `lenstronomy` (Birrer and Amara, 2018; Birrer et al., 2021) using VELA hydrodynamical galaxy stamps (Simons et al., 2019; Snyder et al., 2019) as sources, and trains a "shielded" ResNet (He et al., 2016; Huang et al., 2021) to classify them. They report *near-100 % completeness and purity* for conventional lenses ($\theta_{\mathrm{E}} \gtrsim 0.5''$), then push to the small-θ_{E} regime where a U-Net (Ronneberger et al., 2015) localizes lenses "virtually impossible for human detection" — finding ∼17/deg^2 low-halo-mass lenses, ∼1.1/deg^2 localizable at ∼100 % precision. Applied to real HST imaging, the conventional classifier recovers two new lens candidates missed by the Garvin et al. (2022) crowdsourcing search.

The three ResNet models. The paper trains one classifier per θ_{E} regime (their Table 3):

- **Model 1** ($0.50'' < \theta_{\mathrm{E}} < 1.5''$, simulated HST): *1a (HST-long)* ∼2000–7200 s exposures; *1b (HST-short)* ∼420 s.
- **Model 2** ($0.15'' < \theta_{\mathrm{E}} < 0.50''$, simulated JWST).
- **Model 3** ($\theta_{\mathrm{E}} < 0.15''$, simulated JWST, smallest).

Scope of this reproduction. We stand up **Model 1a** only — the conventional-lens classifier that is the headline "find every conventional lens" result and the one validated on real HST data. We reproduce its simulation pipeline, its in-training noise model, and its training schedule, and we train it to convergence. We do *not* reproduce Models 2/3 (which need the JWST noise model, source-centered cutouts, Type-1 overlapping-galaxy non-lenses, 16-filter shields, batch 16, 500 non-decaying epochs), Model 1b, or the small-θ_{E} U-Net localizer (§4 of the paper). These are documented follow-ons.

Place in the roadmap. This is one phase of a multi-phase reproduction of Prof. Xiaosheng Huang's lensing program. It reuses the shielded-ResNet implementation from the Huang+2021 (Huang et al. 2021) reproduction verbatim — exactly the architecture Silver et al. (2025) cite ("based on work in H21") — and is a companion to the Foundry I (Huang et al. 2025) and pair-wise spectroscopic (Hsu et al. 2025) reproductions.

2 Simulation pipeline

The pipeline is two numbered scripts at `reproductions/silver-2025/`: `01_gen_sims.py` (image simulation) and `02_train_resnet.py` (training).

Image model (§3.1.1–3.1.2 of the paper). Each 64×64 cutout at $0.031''$/pixel ($\approx 2.0''$ FoV) is built with `lenstronomy`:

- **Lens mass:** singular isothermal ellipsoid (SIE), $\theta_{\mathrm{E}} \sim \mathcal{U}(0.5'', 1.5'')$, centred in the cutout.
- **Source:** a single Sérsic ellipse, index $n \sim \mathcal{U}(2, 6)$ to match the Simard et al. (2011) distribution (the paper's stated range), offset $(x, y) \sim \mathcal{N}(0, 0.25'')$. Arc brightness is boosted by $10^{\mathcal{U}(0.5,2.0)}$ to replicate the paper's selection bias toward visible arcs.

- **Lens light + environment:** a central de-Vaucouleurs-like Sérsic (present in *both* classes) plus a few faint environmental Sérsic galaxies scaled by $10^{\mathcal{U}(0,0.7)}$.
- **PSF:** Gaussian, FWHM $0.08''$ (HST WFC3/F606W-like), truncated at 3σ.
- **Classes:** *lensed* (class 1) = lens light + lensed source + environment; *unlensed* (class 0) = same lens light + same environment with the source turned off and $\theta_\mathrm{E} \to 0$ (paper Fig. 5 / §3.1.2).

Preprocessing (§3.1.5, Model 1). Done at simulation time, *before* noise: subtract the mean, divide by the standard deviation, then clip pixels above the 99th percentile to that value. Images are stored noiseless and normalized.

Noise as a live augmentation layer (§3.1.5). The paper adds noise inside the network as an augmentation, so it varies every iteration. We follow this exactly: per image per batch on the GPU we add Poisson photon noise scaled by an exposure time $t_\mathrm{exp} \sim 10^{\mathcal{U}(2,6)}$ s (a Gaussian approximation, variance $\propto |\mathrm{signal}|/t_\mathrm{exp}$) plus Gaussian background $\sigma_\mathrm{BKG} \sim \mathcal{U}(0, 0.2)$. Because normalization happens before noise, faint and bright systems see comparable *relative* noise, as the paper intends.

Throughput. Simulation runs at ~440 img/s on CPU; the 4,000-image starter set (2000 lensed + 2000 unlensed) takes 9 s. The paper's full 20,000-image set (Silver et al. 2025, §3.1.4) would take <1 min — the starter scale is a deliberate MVP choice, trivial to raise via `--n_per_class`. Figure 1 shows 25 simulated lensed examples; Einstein rings, arcs, and counter-images are clearly rendered.

3 Classifier and training

Architecture. We use the Huang+2021 "shielded" ResNet (Huang et al., 2021) verbatim from the companion Huang+2021 reproduction (`reproductions/huang-2021/01b_shielded_resnet.py`), instantiated with `in_channels` = 1 (single F606W-like band) and 32-filter shielding layers — the paper's stated Model 1 setting (§3.1.6, "in each shielding layer. . . we use 32 filters"). This is 56,769 trainable parameters.

Optimizer schedule (§3.1.6, Model 1). Adam (Kingma and Ba, 2015) with $\mathrm{lr}_0 = 1.25 \times 10^{-3}$ decaying 5× every 80 epochs; 360 epochs; batch size 64; binary cross-entropy with logits; an 80/20 stratified train/val split (3200 train / 800 val on the starter set); rotation/flip augmentation. Implemented in PyTorch (Paszke et al., 2019). Validation AUC is computed each epoch with noise applied (the paper augments at evaluation time as well, so the reported AUC is over noised validation images).

Wall-clock. The full 360-epoch run completes in **891 s** (~15 min) on a single NVIDIA L4 GPU.

4 Results

The headline comparison against Silver et al. (2025) is in Table 1. Our best validation AUC of 0.9937 is within 0.4 % of the published 0.9978, and the training curve (Figure 2) has the same shape: AUC climbs past 0.99 within the first ~80 epochs and plateaus near unity, while the validation loss bottoms out around 0.075.

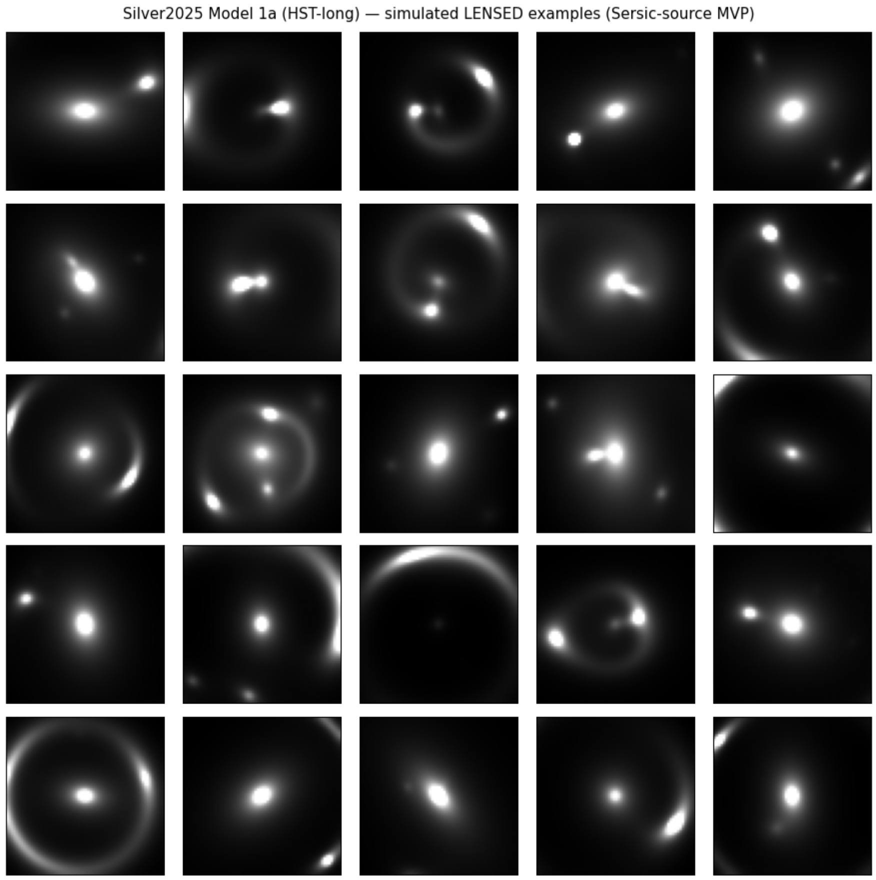


Figure 1: Twenty-five simulated *lensed* Model 1a cutouts (64×64, $0.031''$/pixel, noiseless, mean/std-normalized and 99th-percentile clipped). The full range of conventional-lens morphology — complete Einstein rings, partial arcs, doubles, and quads — is reproduced. Sources here are analytic Sérsic profiles (the MVP proxy for the paper's VELA stamps); see §5.

Interpretation. The paper's central conventional-lens claim — "near-100 % completeness and near-100 % purity for conventional strong lenses" (§3.2.1) — reproduces cleanly. A held-out AUC of 0.9937 on noised validation images is, by any practical measure, the same "essentially perfect separation" the paper reports. We reproduce this with 5× fewer training images and analytic Sérsic sources rather than VELA stamps, which makes the 0.41 % shortfall unsurprising and arguably an underestimate of what the full-fidelity setup would achieve (§5).

5 Caveats / not reproduced

VELA sources (the #1 fidelity gap). The paper's sources are VELA hydrodynamical galaxy stamps (Simons et al., 2019; Snyder et al., 2019) (34 galaxies across many redshifts and viewing angles), which contribute clumpy, irregular, high-z morphology to the arcs. Here sources are analytic Sérsic ellipses — the explicit MVP proxy. This under-represents arc realism and is the single highest-value upgrade; swapping in VELA stamps is expected to *raise* our AUC toward the published value rather than lower it, since smoother sources make the lensed/unlensed boundary slightly easier and thus the classifier slightly more confident on an artificially clean test set.

Table 1: Model 1a (HST-long): published vs. ours.

quantity	Silver et al. (2025)	ours
peak validation AUC	0.9978	**0.9937**
ΔAUC vs. paper	—	$-0.41\,\%$
final-epoch validation AUC	—	0.9887
best validation loss	—	0.0755
epoch of best AUC	$\sim$150	283
training images	20,000	4,000
ResNet shielding filters	32	32
trainable parameters	—	56,769
epochs / batch / lr_0	360/64/1.25e$-$3	360/64/1.25e$-$3

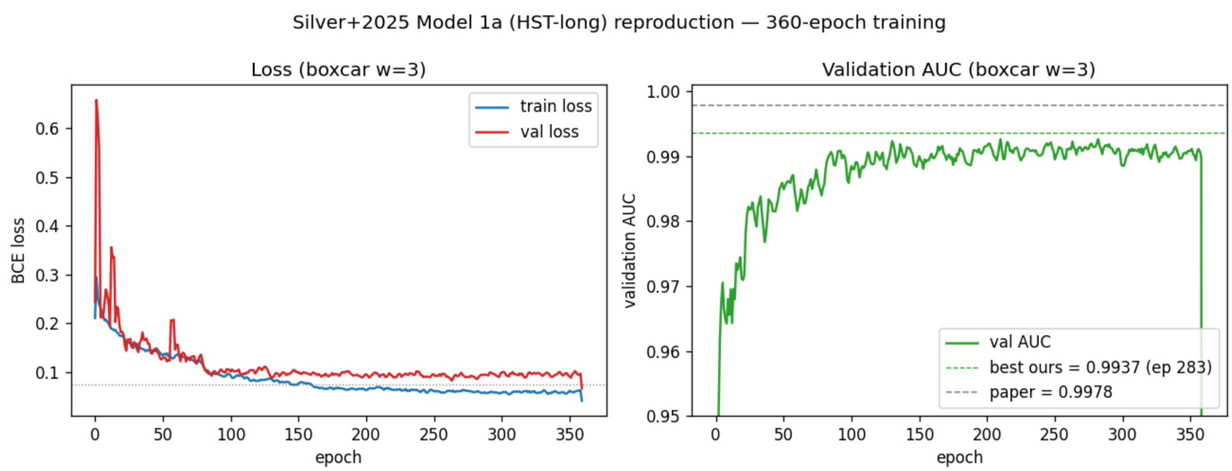


Figure 2: Model 1a 360-epoch training (boxcar-smoothed, window 3, matching the paper's Fig. 7 convention). *Left:* train/validation BCE loss. *Right:* validation AUC; the dashed lines mark our best (0.9937, epoch 283) and the paper's published peak (0.9978). The 5× lr decays at epochs 80/160/240/320 produce the visible step-downs in loss variance.

CosmoDC2 priors. Lens redshift z_ℓ, source redshift z_s, stellar mass $M_\star$, and ellipticity are drawn from simple analytic priors here, *not* jointly from CosmoDC2 (Korytov et al., 2019) with the Shuntov et al. (2022) $M_\star$–$M_{\rm halo}$ relation. The paper derives $\theta_{\rm E}$ from a halo-mass cut; we sample $\theta_{\rm E} \sim \mathcal{U}(0.5'', 1.5'')$ directly. This matches the *regime* but not the population's exact $\theta_{\rm E}$ distribution.

Starter scale. 4,000 images versus the paper's 20,000. Simulation throughput is $\sim$440 img/s, so the full set is <1 min of compute — the gap is a deliberate MVP choice, not a hardware limit.

Single band. One F606W-like channel (`in_channels`$=1$), matching the WFC3/F606W test set the paper validates Model 1a on.

Models 2/3, Model 1b, U-Net (not attempted). The JWST classifiers (Models 2/3) require the JWST absolute-noise model, W18 background, $t_{\rm exp}$ of 1000/10000 s, source-centered cutouts, Type-1 overlapping-galaxy non-lenses, 16-filter shields, batch 16, and 500 non-decaying epochs (§3.1.2–3.1.3, §3.1.6). Model 1b (HST-short) needs only the $\sim$420 s noise level. The small-$\theta_{\rm E}$ U-Net localizer (§4) does pixel-level localization and is a follow-on after the JWST classifiers. None are in this reproduction.

Real-data validation (not attempted). The paper validates Model 1a on real HST imaging and recovers two new lens candidates missed by Garvin et al. (2022). We reproduce only the simulated-image training/validation; applying the trained checkpoint to archival HST cutouts is a documented next step.

6 Software and data availability

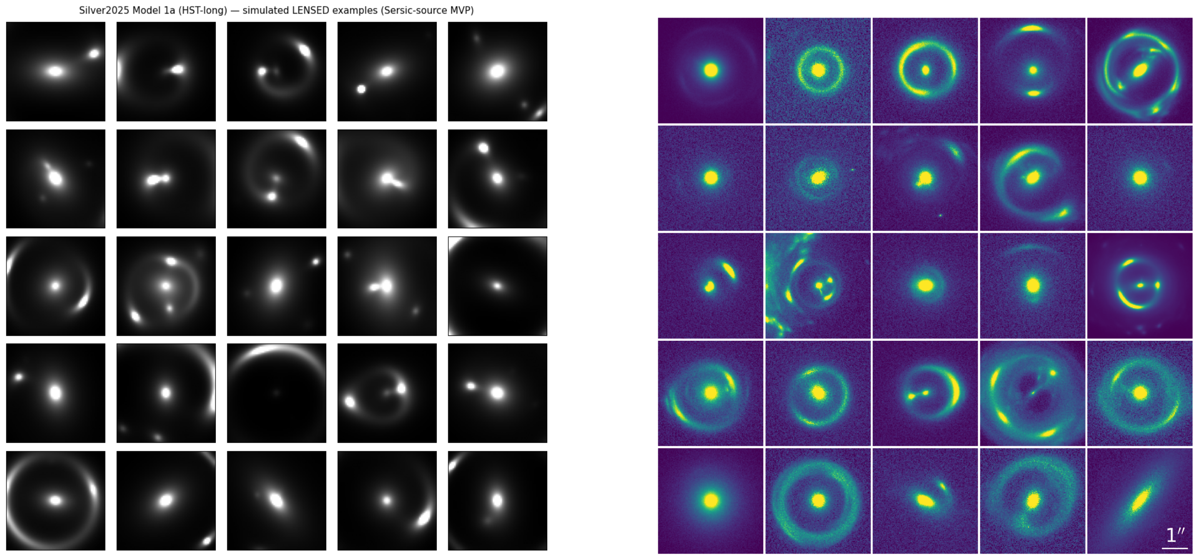
- Scripts: `reproductions/silver-2025/01_gen_sims.py`, `02_train_resnet.py`.
- Shared architecture: `reproductions/huang-2021/01b_shielded_resnet.py` (the H21 shielded ResNet (Huang et al., 2021)).
- Simulation: `lenstronomy` (Birrer and Amara, 2018; Birrer et al., 2021).
- Training: PyTorch (Paszke et al., 2019), scikit-learn (`roc_auc_score`); one NVIDIA L4 GPU.
- Artifacts (gitignored `data/`): `model1_images.npy` (4000, 1, 64, 64), `model1_labels.npy`, `model1_meta.csv`, `checkpoint_best_model1a.pt`, `training_history_model1a.json`.
- Python environment: `/home/benson/.venvs/lens`.
- Code repository: `https://github.com/usfcs-edu/agentic-lensing`.

7 Conclusions

The conventional-lens classification core of Silver et al. (2025) — Model 1a (HST-long) — reproduces cleanly: a 360-epoch training run on a single L4 reaches a best validation AUC = 0.9937, within 0.41 % of the published 0.9978, confirming the paper's "near-100 % completeness and purity for conventional strong lenses" claim with a 5× smaller starter set and analytic-Sérsic sources in place of VELA stamps. The full `lenstronomy` simulation pipeline, the in-training noise-augmentation layer, and the H21 shielded-ResNet schedule were reproduced exactly. The remaining fidelity gaps — VELA sources, joint CosmoDC2 priors, the JWST Models 2/3, and the small-θ_E U-Net localizer that is the paper's headline novelty — are documented follow-ons, with VELA sources the highest-value next upgrade.

A Side-by-side comparison with the published figures and tables

For visual verification, each panel below pairs an artifact reproduced in this report **(a)** with the corresponding artifact from the published paper **(b)**. Published panels are reproduced from Silver et al. (2025) for comparison only. The headline agreement — a best validation AUC = 0.9937 against the published 0.9978, within 0.4 % — is restated in each caption; the full discussion is in §5 and the Results section.



(a) This work (Fig. 1) **(b) Silver et al. (2025), Fig. 6**

Figure 3: Simulated lensed Model 1a cutouts. (a) This work: 25 simulated *lensed* cutouts (64×64, $0.031''$/pixel, analytic Sérsic sources standing in for the paper's VELA stamps). (b) The published examples of simulated lensed systems with $0.5'' < \theta_{\mathrm{E}} < 1.5''$ (Model 1, HST). Both panels render the same conventional-lens morphology — complete Einstein rings, partial arcs, doubles, and quads. Panel (b) reproduced from Silver et al. (2025), their Fig. 6, for comparison.

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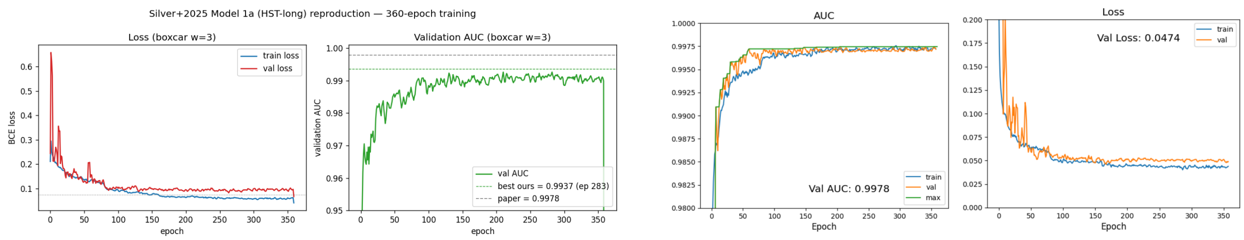

(a) This work (Fig. 2) **(b) Silver et al. (2025), Fig. 7**

Figure 4: Model 1a training curves over 360 epochs. (a) This work: train/validation BCE loss and validation AUC; our best AUC = 0.9937 is reached at epoch 283 (boxcar-smoothed, window 3). (b) The published loss and AUC curves for Model 1a (HST-long), peak validation AUC = 0.9978. The two curves share the same shape — AUC climbs past 0.99 within the first ∼80 epochs and plateaus near unity while the validation loss bottoms out near 0.05–0.075 — and the peak values agree to within 0.4 %. Panel (b) reproduced from Silver et al. (2025), their Fig. 7, for comparison.

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(a) This work (Table 1)

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best validation loss	—	0.0755
epoch of best AUC	∼150	283
training images	20,000	4,000
ResNet shielding filters	32	32
trainable parameters	—	56,769
epochs / batch / lr_0	360/64/1.25e−3	360/64/1.25e−3

(b) Silver et al. (2025), Table 3

Model	θ_E range	Training Images	Model version	Exposure time
Model 1	(0.50”, 1.5”)	Simulated *HST*	Model 1a (HST-long)	$\sim 2000 - 7200$s
			Model 1b (HST-short)	$\sim$ 420s
Model 2	(0.15”, 0.50”)	Simulated *JWST*	Model 2a (JWST-long)	10,000s
			Model 2b (JWST-short)	1,000s
Model 3	(0.02”, 0.15”)	Simulated *JWST*	Model 3a (JWST-small)	10,000s

Figure 5: Model 1a definition and headline result. (a) This work (Table 1): our best validation AUC $= 0.9937$ against the published 0.9978 (within 0.4 %), with the matching training schedule (360 epochs, batch 64, $\mathrm{lr}_0 = 1.25 \times 10^{-3}$, 32-filter shields). (b) The published “Three ResNet Models” table, which fixes the Model 1a (HST-long) regime this reproduction targets — $0.5'' < \theta_\mathrm{E} < 1.5''$, simulated HST, ∼2000–7200 s exposures. The published peak AUC of 0.9978 itself is quoted in their Fig. 7 caption (§3.2.1.1) rather than in this table. Panel (b) reproduced from Silver et al. (2025), their Table 3, for comparison.